

AI Ethics

GLOSSARY 06 - SHEET 1 / 8

The applied field concerned with the moral acceptability of designing and deploying AI

PROFESSIONAL DEFINITION

AI ethics is the field of applied ethics concerned with the moral acceptability of the design, development, deployment and use of artificial intelligence systems, and with the obligations of those who commission and operate them. In the European context it is operationalised through the four principles of the Ethics Guidelines for Trustworthy AI - respect for human autonomy, prevention of harm, fairness and explicability - and their seven implementing requirements (HLEG AI, 2019; Floridi and Cowls, 2019).

WHO DEFINED IT ORIGINALLY

The four-principle formulation was adopted by the European Commission's High-Level Expert Group on Artificial Intelligence in 2019. Luciano Floridi of the University of Oxford and Josh Cowls (2019) derived a closely related unified set of five principles from a comparative analysis of six major ethics declarations, adding explicability to the four principles of biomedical ethics established by Beauchamp and Childress (1979).

WHY THIS DEFINITION WAS DEVELOPED

By 2018 dozens of organisations had published AI ethics principles, with substantial overlap and no common vocabulary, which made compliance claims incomparable. Floridi and Cowls demonstrated the convergence and supplied a single set; the High-Level Expert Group then attached implementing requirements, converting principles into an assessable form.

FIGURE 1 - FOUR PRINCIPLES, SEVEN REQUIREMENTS

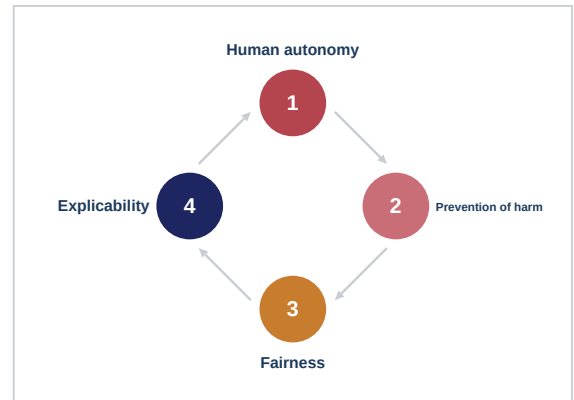


Figure 1. The four principles are not independently satisfiable. Explicability requirements can conflict with harm prevention where disclosure creates risk, and the auditable artefact is the recorded reasoning about how the conflict was resolved (after HLEG AI, 2019; Floridi and Cowls, 2019).

HOW IT IS USED

Apply the seven requirements as an assessment instrument during design rather than as a review at release, and record the trade-offs made between principles, since the principles conflict in practice and the record of how conflict was resolved is the auditable artefact.

WHEN IT IS USED

At the design stage of any consequential AI system, at ethics review board submission, in vendor assessment, and where two principles cannot be simultaneously satisfied.

BY WHOM IT IS USED

AI ethics boards and review committees, product owners and designers, risk and compliance functions, and executives accountable for deployment decisions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor instruments were general corporate codes of conduct, which were addressed to employee behaviour and gave designers no guidance about system properties. The trustworthy AI framework specifies requirements at the level of the system - human agency, technical robustness, privacy, transparency, fairness, societal wellbeing, accountability - and pairs them with an assessment list. Its acknowledged limitation is that it remains voluntary; the AI Act converts a subset into binding obligation, but the ethical requirements exceed the legal ones.

RELATED DEFINITIONS IN THIS COURSE

[Trustworthy AI - Unit 6](#)
[EU AI Act - Unit 6](#)
[Algorithmic Bias - Unit 6](#)
[Accountability Gap - Unit 6](#)
[Externality - Unit 5](#)

SOURCES (HARVARD REFERENCING STYLE)

Beauchamp, T.L. and Childress, J.F. (1979) *Principles of Biomedical Ethics*. New York: Oxford University Press.
 Floridi, L. and Cowls, J. (2019) 'A unified framework of five principles for AI in society', *Harvard Data Science Review*, 1(1).
 High-Level Expert Group on Artificial Intelligence (2019) *Ethics Guidelines for Trustworthy AI*. Brussels: European Commission.
 Jobin, A., Ienca, M. and Vayena, E. (2019) 'The global landscape of AI ethics guidelines', *Nature Machine Intelligence*, 1(9), pp. 389-399.